

AI Usage Report

1. AI Usage Declaration

During PA4-2026, our team used AI tools to support software development and documentation activities.

The team remains responsible for reviewing, validating, and integrating all AI-assisted outputs into the project.

2. AI Tools Used

AI Tool	Purpose
ChatGPT	Software architecture documentation, Use-Case refinement, text formatting, and Mermaid diagram generation
GitHub Copilot	Code assistance, autocomplete, and development support for frontend/backend implementation
Spec Kit	AI-driven specification, task breakdown, and test case generation for new functional groups

3. AI Usage Log

Date	Team Member	AI Tool	Purpose	Output / Result
29/07/2026	Nguyễn Tường Huy	ChatGPT	Revise Use-Case Model and Specs	Improved clarity and concise behavioral descriptions based on PA3 feedback
01/08/2026	Nguyễn Tường Huy	ChatGPT	Draft C4 Level 1 & Level 3 (Frontend) diagrams	Generated initial Mermaid syntax for System Context and Frontend Components
01/08/2026	Nguyễn Công Chiến	ChatGPT	Draft C4 Level 2 & Level 3 (Backend) diagrams	Generated initial Mermaid syntax for Container and Backend Components
02/08/2026	Ngô Trung Nghĩa	ChatGPT	Deployment Diagram and text formatting	Drafted Deployment Diagram in Mermaid and checked grammar for Changes.md and Weekly Report
03/08/2026	Lê Hoàng Tấn, Nguyễn Minh Hoàng	Spec Kit	Generate artifacts for 2 new functional groups	Generated specs, plans, and task breakdowns for FG1 and FG2
04/08/2026	Nguyễn Minh Hoàng, Lê Hoàng Tấn	GitHub Copilot	Backend API and UI implementation	Faster generation of React components and Express API boilerplate for FG1 and FG2

Date	Team Member	AI Tool	Purpose	Output / Result
05/08/2026	Vũ Lê Trọng Văn	Spec Kit	Generate test cases	Automated generation of Spec Kit test cases for the newly implemented functional groups

4. AI-Assisted Activities

AI tools were used for activities such as:

- **Documentation & Requirements:** Refining Use-Case Specifications and Model (Huy), tracking changes, and polishing the Weekly Report (Nghĩa).
- **Architecture Design:** Generating and structuring Mermaid code for System Context & Frontend (Huy), Container & Backend (Chiến), and Deployment (Nghĩa) architecture diagrams.
- **Feature Specification:** Using Spec Kit to generate formal specifications and task breakdowns for 2 new functional groups (Hoàng & Tấn).
- **Implementation:** Assisting with code-related questions and writing frontend/backend components using GitHub Copilot (Hoàng & Tấn).
- **Testing:** Generating Spec Kit test cases for the implemented features (Văn).

5. Human Review

All AI-generated content was reviewed by team members before being included in the project.

The team verified that:

- The generated information is relevant to the project and aligns with PA4 requirements.
- Architecture diagrams and deployment mappings exactly match the actual source code implementation.
- Code suggestions were thoroughly tested, reviewed, and integrated manually.
- Final architectural, documentation, and technical decisions were made by the project team.

6. Responsibility

AI tools were used strictly as assistance rather than as a replacement for the team's development, analysis, and decision-making process.

The project team maintains full ownership and is responsible for the final source code, documentation, architecture, and implementation.